

Automatic layer-change routing

A step-by-step guide to assigning track width and differential-pair gap by copper layer, applying a reviewed profile, and routing with Constraint Studio closed.

What changes automatically?

After the review is applied and the project is reopened, KiCad chooses the saved width and pair gap when a **new** route starts on each configured layer. Studio is not a background routing hook.

Save board > Set profile > Stage and export > Apply and reopen > Route

At a transition via: KiCad 10 keeps one width during an active route. Finish at the via, start a new route on the destination layer, and turn **Auto track width off**. Otherwise KiCad can inherit the previous segment's width.

All dimensions shown in the screenshots are illustrative. Use values approved for your actual stackup and fabricator.

1 Save the board and physical stackup

1. Open the project in KiCad PCB Editor. Verify enabled copper layers, dielectrics and copper thicknesses in **Board Setup**.
2. Save Board Setup and the PCB. Confirm the nets to be routed are present in the saved board.
3. Open **Tools > External Plugins > WayriCAD Constraint Studio**. If the board changed while Studio was open, return to a clean workspace or use **Reload saved board**. Export or undo staged edits first.

Check: Physical stackup shows the expected layers. A missing net usually indicates a stale saved snapshot or a net not yet present on the PCB.

2 Create a profile for each protocol instance

1. Open **Layer routing profiles** and choose **New profile** or **+ New protocol** (label varies by build).
2. Name the specific bus, such as CAN1. Choose its protocol family and **Signal type**.
3. Under **Apply to**, choose selected nets/group or an existing netclass. Search and tick exact saved-board nets. For a pair, include both members, for example CAN_P and CAN_N. The family label alone does not create a KiCad pair.
4. Create separate tabs for CAN2, Ethernet ports or DDR groups with different geometry. Duplicate copies geometry but clears the net selection.

WayriCAD Constraint Studio

workflow_demo.kicad_pcb
C:/Projects/Example/workflow_demo.kicad_pcb

Unreported changes | Change project | Export review

DESIGN INTENT / VERIFIED RULES

Constraint worksheet

Clearance matrices

Layer routing profiles

Netclasses & settings

Constraint sets

Review & export

Engineering tools

Reference & help

Local workspace
Dragged with: Your source stays intact.

Theme Help / F1

WayriCAD 3.1.1 Development build

PROTOCOL - STACKUP - NATIVE ROUTER

Layer routing profiles

Set geometry once. KiCad selects the saved width and pair gap as you change routing layers.

After applying the review and reopening the project, use KiCad's netclass/rule sizing mode. No manual width or gap selection is needed at each layer change. Studio can be closed. Existing copper is not resized.

How it works | New profile | Review & stage profile

Physical stackup Saved board

F.Cu
0.035 mm
copper

dielectric 1
0.2 mm - er 4.2
prepreg

In1.Cu
0.035 mm
copper

dielectric 2
1.0 mm - er 4.2
core

In2.Cu
0.035 mm
copper

dielectric 3
0.3 mm - er 4.2
prepreg

B.Cu
0.035 mm
copper

References must be continuous planes along the route. A named copper layer alone does not establish that continuity. Via transitions also need a return-path review.

Routing intent

Instance / group name
CAN1

Protocol family
CAN

Apply to
Selected nets / group
2 exact nets selected - no netclass reassignment

Signal type
differential

Target impedance (Ω)
120

Width / gap DRC tolerance (%)
5

☒ Prohibit tracks on unconfigured layers

Net membership - 2 selected

Filter nets, e.g. CAN1 or D

Select visible | Clear visible

☒ CAN1_N ☒ CAN1_P ☐ CAN2_N **CAN2** ☐ CAN2_P **CAN2**

☐ D10 ☐ D11 ☐ D12 ☐ D13

☐ D14 ☐ D15 ☐ D16 ☐ DDR1_DQ0 **DDR1 Data** ☐ DDR1_DQ1 **DDR1 Data** ☐ DDR1_DQ2 **DDR1 Data** ☐ ETH1_TX_N **Ethernet1 TX** ☐ ETH1_TX_P **Ethernet1 TX** ☐ ETH1_TX_Q **Ethernet1 TX**

Selected: CAN1_N, CAN1_P

Exact names only; the filter selects existing nets, not a wildcard rule. New or renamed nets require membership review. Differential groups must include both pair members.

Use separate instances for each CAN bus, Ethernet port, or DDR data/address/clock group. A group shares one signal type and layer geometry. For pair routing, KiCad must recognize the two nets as a differential pair. The percentage above is a geometry limit, not an impedance tolerance. Existing scoped custom rules retain higher priority; review their overrides in the worksheet.

Per-layer geometry + Signal layer

Signal layer	Reference 1	Upper reference (stripline)	Width mm	Pair gap mm	
F.Cu	In1.Cu	None — outer l	0.2	0.18	Estimate width x
B.Cu	In2.Cu	None — outer l	0.3	0.25	Estimate width x

Enter fabricator/field-solver dimensions, or estimate width with a fixed pair gap. Estimates support homogeneous microstrip and symmetric stripline only; they omit soldermask and conductor roughness. Review every value before staging.

F.Cu → 0.2 mm width / 0.18 mm gap Reference: In1.Cu

B.Cu → 0.3 mm width / 0.25 mm gap Reference: In2.Cu

Protocol instances & native profiles

Managed profiles can be reopened here. Other native profiles are preserved.

Instance	Protocol family	2 nets - 2 layers - 120 Ω	Edit instance	Duplicate	Remove
CAN1	CAN	2 nets - 2 layers - 120 Ω	Edit instance	Duplicate	Remove
CAN2	CAN	2 nets - 2 layers - 120 Ω	Edit instance	Duplicate	Remove
DDR1 Data	DDR	3 nets - 2 layers - 50 Ω	Edit instance	Duplicate	Remove
Ethernet1 TX	Ethernet	2 nets - 2 layers - 100 Ω	Edit instance	Duplicate	Remove

Local lint and scope previews are separate from native DRC. Ctrl+Z Undo - Ctrl+Y Redo - / Search

Figure 1. Source-derived rendering of the shipped interface, using demonstration values. It is not a PCB Editor screenshot.

Select the exact nets

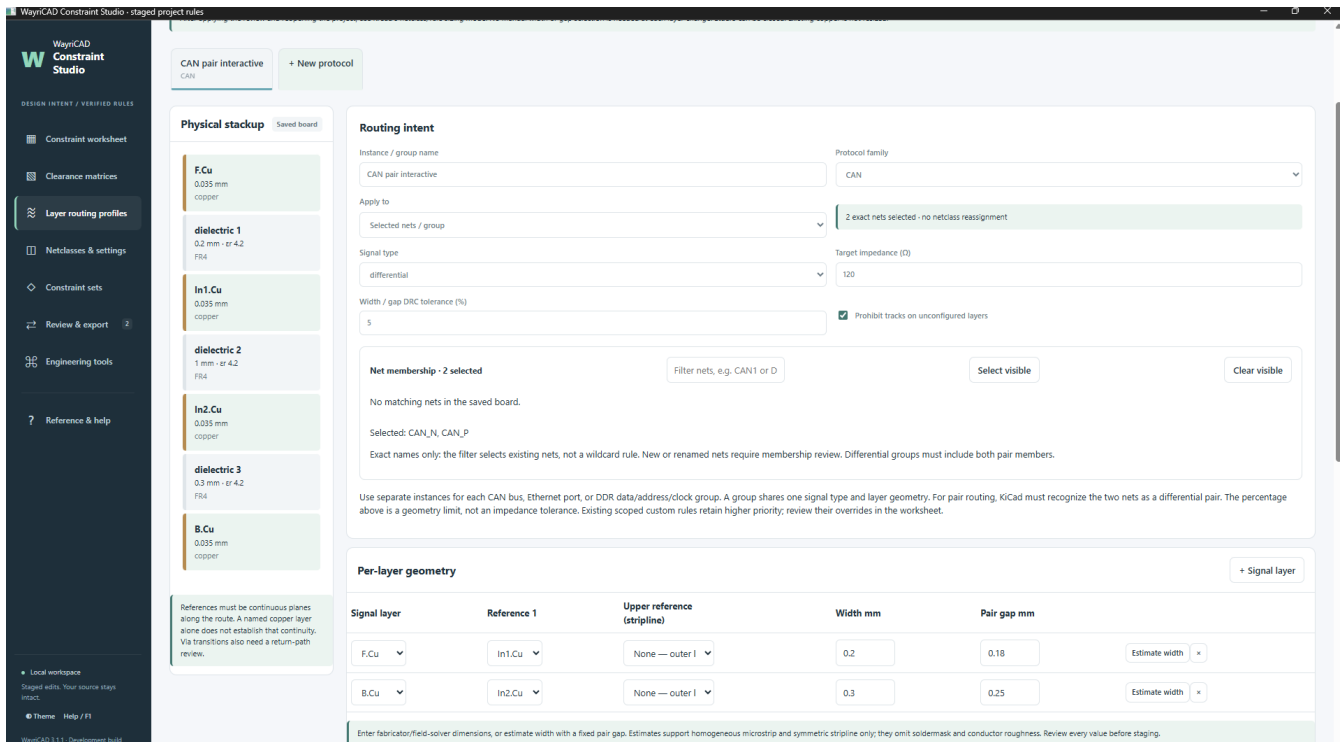


Figure 2. User-provided Constraint Studio screenshot. If the list says 'No matching nets,' reload the saved board before staging.

3 Enter geometry for every permitted layer

1. Click **+ Signal layer** for each copper layer this group may use. Select the signal layer and its actual reference plane or planes.
2. Enter **Width mm** and, for a pair, **Pair gap mm** on each row. The pair gap is edge-to-edge. Use verified stackup or fabricator values; Estimate width is only a screening approximation at a fixed gap.
3. Compare the shared-scale **Layer previews**. Trace, Pair and Trace + pair modes should update visibly as dimensions change.
4. Optionally **Prohibit tracks on unconfigured layers**. This policy does not prohibit vias.

Illustrative layer	Width	Pair gap	New route uses
F.Cu	0.20 mm	0.18 mm	Front-layer geometry
B.Cu	0.30 mm	0.25 mm	Back-layer geometry

Compare the layer previews

In2.Cu
0.035 mm
copper

dielectric 3
0.3 mm - er 4.2
FR4

B.Cu
0.035 mm
copper

References must be continuous planes along the route. A named copper layer alone does not establish that continuity. Via transitions also need a return-path review.

No matching nets in the saved board.

Selected: CAN_N, CAN_P

Exact names only: the filter selects existing nets, not a wildcard rule. New or renamed nets require membership review. Differential groups must include both pair members.

Use separate instances for each CAN bus, Ethernet port, or DDR data/address/clock group. A group shares one signal type and layer geometry. For pair routing, KiCad must recognize the two net above is a geometry limit, not an impedance tolerance. Existing scoped custom rules retain higher priority; review their overrides in the worksheet.

Per-layer geometry

Signal layer	Reference 1	Upper reference (stripline)	Width mm	Pair gap mm	
F.Cu	In1.Cu	None — outer I	0.2	0.18	E
B.Cu	In2.Cu	None — outer I	0.3	0.25	E

Enter fabricator/field-solver dimensions, or estimate width with a fixed pair gap. Estimates support homogeneous microstrip and symmetric stripline only; they omit soldermask and conductor roughness. Review every value before s

Layer previews
Same scale across this instance - vertical copper geometry

F.Cu



Width: 0.2 mm
Pair gap: 0.18 mm

B.Cu



Width: 0.3 mm
Pair gap: 0.25 mm

Figure 3. User-provided shared-scale layer preview. It illustrates geometry; it does not calculate impedance or alter PCB copper.

Reference-plane check: A named copper layer does not prove an unbroken return plane. Inspect continuity, plane voids and the return path at each via.

4 Review the optional via recipe

If the bus needs a particular via, choose through, microvia, blind or buried type, copper diameter and drill. Set a valid start/end copper-layer span for a non-through via. **Use saved sizes** copies netclass dimensions for review. Board Setup must permit the via type and the recipe must meet manufacturing floors.

Studio stages scoped DRC limits for the recipe. KiCad's tuning profile does **not** automatically switch the router's via type; select it in PCB Editor.

5 Stage, review, export and apply

1. Click **Review & stage profile**. Resolve missing nets, overlapping assignments, stale stackup and invalid dimensions.
2. In **Review & export**, inspect Local findings, File diff and Native rules. Export to a new empty folder, then run native KiCad DRC on that exported snapshot.
3. Click **Open Apply Review**. Close every source-project KiCad editor before confirming the helper's write. Review its file list; it checks fingerprints and creates backups.
4. Reopen the project in PCB Editor. The profile persists with Studio closed. If the stackup changes, reopen Studio and restage.

Stage alone changes only the Studio workspace. The open PCB Editor does not receive the new rules until the reviewed export is applied and the project is reopened.

6 Route from one layer to another

1. Use KiCad **netclass/rule-driven sizing** for tracks or pairs. Clear explicit manual width/gap overrides. Turn **Auto track width off**.
2. Start a new route on the first configured layer and inspect KiCad's width and, for a pair, gap.
3. Place the permitted transition via, then **finish the active route at the via**.
4. Switch layers and start a **new** route from the via. Inspect the destination layer's width and gap. Repeat for every transition.
5. Save the PCB, run native DRC, and review each pair member and return-path transition.

Illustration only: a new F.Cu route may use 0.20 mm width / 0.18 mm pair gap, and a new B.Cu route may use 0.30 mm / 0.25 mm after restarting at the via.

Verify that the profile is working

- Start a new route on every configured layer and inspect the chosen width and gap.
- Close and reopen the project, then repeat with Studio closed to confirm persistence.
- Inspect saved track properties and run DRC. Existing copper is not resized.
- Test a net from each CAN, Ethernet or DDR instance; new net names are not silently included.

The recorded KiCad 10.0.6 single-track check measured 0.20 mm on F.Cu and 0.30 mm on B.Cu after a route restart with Auto track width off. The user subsequently confirmed differential-pair layer routing in their session; the repository has no measured route capture for that pair test.

If a size does not change

Symptom	Check
Net list empty or stale	Save the PCB, return to a clean Studio workspace and Reload saved board. Confirm both pair nets exist on the PCB.
New layer keeps old width	Turn Auto track width off; finish at the via and start a new route. Clear manual overrides.
Wrong pair gap	Check KiCad-recognized pair names, both selected members, destination-layer gap row and higher-priority scoped rules.
Wrong via type	Select the type in the router. Studio's via recipe is a DRC constraint, not automatic selection.
Changes absent after Stage	Export and apply the review with source editors closed, then reopen the project.
Profile marked stale	Review the changed saved stackup or owned rules and restage.

Screenshot provenance: Figure 1 is a source-derived UI rendering with demo data. Figures 2-3 are user-provided Constraint Studio captures. None is a fabrication or impedance sign-off.